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SATURATION SPECTROSCOPY WITH RUBIDIUM – 29.10.2025

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1 Aim of the experiment

The aim of this experiment is to record and analyze both the Doppler-broadened and Doppler-free absorption spectra of rubidium using a grating-stabilized diode laser at 780 nm. In the first part, the Doppler-broadened spectrum is measured to determine the thermal velocity distribution and the corresponding Doppler width of the atomic ensemble. In the second part, Doppler-free saturation spectroscopy is performed in a pump–probe configuration to resolve the fine and hyperfine structure of the rubidium D₂ transition, revealing narrow Lamb dips and crossover resonances. From these spectra, the frequency calibration, atomic temperature, and characteristic linewidths are determined, providing a fundamental understanding of high-resolution laser spectroscopy and laser frequency stabilization.

2 Preparatory Problems

In the following section, several fundamental theoretical aspects related to the experiment are discussed. The questions listed below summarize the key physical concepts required to understand the measurements performed in the laboratory.

- 1. Why is the width of the velocity distribution in a thermal ensemble of atoms proportional to $\sqrt{kT/M}$?**

The atomic velocities follow the Maxwell–Boltzmann distribution, which has a Gaussian shape with standard deviation:

$$\sigma_v = \sqrt{\frac{kT}{M}}.$$

- 2. We use in this experiment a laser with a wavelength of 780 nm. To which values does this correspond in Hz and eV?**

The wavelength corresponds to:

$$\nu = \frac{c}{\lambda} \approx 3.84 \times 10^{14} \text{ Hz} \quad \text{and} \quad E = h\nu \approx 1.59 \text{ eV}.$$

3. The laser frequency of the external cavity diode laser is changed by modulating the cavity length using a piezoelectric crystal. How does the laser frequency depend on the cavity length?

For the resonance condition:

$$m\lambda = 2L$$

the laser frequency is

$$\nu = \frac{mc}{2L}$$

and thus

$$\frac{d\nu}{dL} = -\frac{\nu}{L}$$

meaning an increase in L decreases ν .

4. Write down the manifolds of atomic states of rubidium-87 or rubidium-85 that exist due to fine, hyperfine, and Zeeman substructure. Please list the possible internal states of the atom in the Dirac notation $|F, m_F\rangle$.

Rubidium-85 and rubidium-87 both show fine and hyperfine splitting due to the interaction between the electron angular momentum and the nuclear spin.

$$Rb^{87}: \quad I = \frac{3}{2}, \quad 5^2S_{1/2}: F = 1, 2, \quad 5^2P_{3/2}: F' = 0, 1, 2, 3$$

$$Rb^{85}: \quad I = \frac{5}{2}, \quad 5^2S_{1/2}: F = 2, 3, \quad 5^2P_{3/2}: F' = 1, 2, 3, 4$$

$$|F, m_F\rangle \quad \text{with} \quad m_F = -F, \dots, +F$$

5. Which transitions in the atomic structure of rubidium are allowed according to the selection rules?

Allowed transitions satisfy:

$$\Delta F = 0, \pm 1 \quad (\text{but not } F = 0 \leftrightarrow F' = 0)$$

$$\Delta m_F = 0, \pm 1$$

6. When performing saturation spectroscopy, how many resonance peaks do you expect for a given rubidium isotope and a given fine-structure transition?

For example, in rubidium-87 with the ground state $F = 2$, transitions to the excited states $F' = 1, 2$, and 3 are possible, giving three main peaks. A crossover peak occurs between each pair of transitions, resulting in a total of six peaks. This number is the same for all isotopes and fine-structure transitions.

7. Why can crossover resonances be observed in the frequency spectrum, and where are they located in the energy-level diagram?

Crossover resonances occur when atoms moving with a specific velocity interact simultaneously with both the pump and the probe beam. In this case, the pump beam is resonant with one transition (ν_1) and the probe beam with another (ν_2) due to opposite Doppler shifts. This leads to a reduced absorption for the probe beam, producing an additional peak in the transmission spectrum. These crossover peaks appear exactly halfway between the two corresponding hyperfine transitions at the frequency:

$$\nu_{\text{co}} = \frac{\nu_1 + \nu_2}{2}$$

3 Data analysis

In this section, the recorded data from the rubidium spectroscopy experiment are processed and evaluated. The analysis includes the determination of the frequency–voltage calibration factor, identification and fitting of the Doppler-broadened and Doppler-free absorption spectra, and the extraction of relevant physical parameters such as the Doppler width, temperature, and frequency separations between hyperfine transitions. All evaluations were performed using Python to ensure a consistent and quantitative analysis of the measured data.

3.1 Determination of the ratio $\frac{d\nu}{dV}$:

Finding the proportionality factor between the applied ramp voltage V (used to scan the laser frequency) and the corresponding change in optical frequency ν is the aim of this step.

The voltage locations of the two isotope absorption dips in the Doppler-broadened rubidium spectrum must be determined in order to accomplish this. The measured CSV data was used to plot the spectrum in Python, and a numerical minimum search was used to automatically identify the dip locations from the plotted curve. The plots are shown in *Figure 1* and 2.

The two main absorption minima correspond to the transitions

$$^{87}\text{Rb} (5S_{1/2}, F = 2) \rightarrow (5P_{3/2}, F' = 1, 2, 3)$$

and

$$^{85}\text{Rb} (5S_{1/2}, F=3) \rightarrow (5P_{3/2}, F' = 2, 3, 4).$$

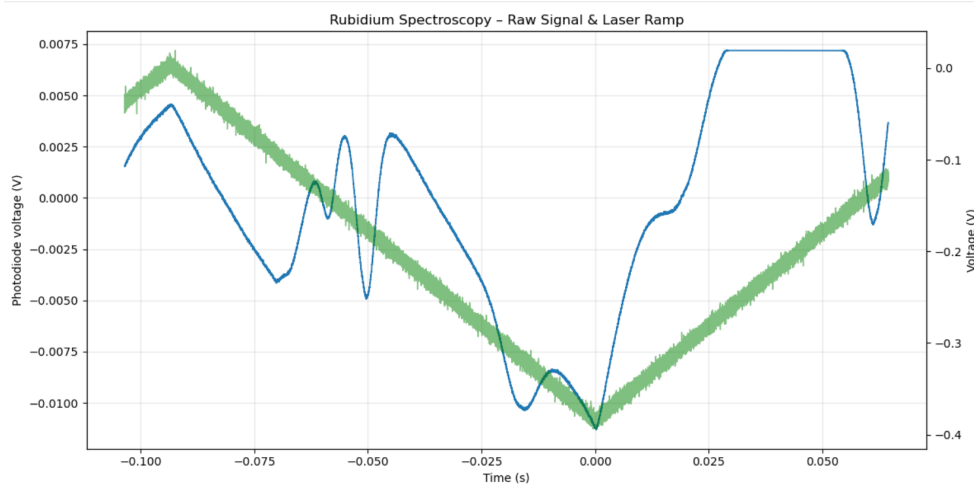


Figure 1 Plot of the signal of the photodiode

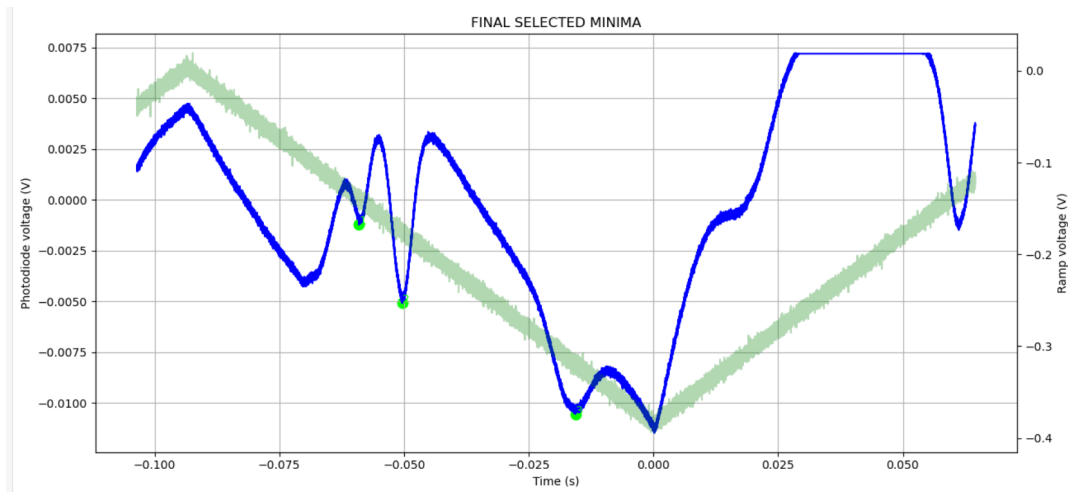


Figure 2 Plot with the calculated dips

From the Python analysis, the following voltages for the three found dips were obtained:

$$V_1 = -0.142676\text{V } (^{87}\text{Rb } F=2),$$

$$V_2 = -0.175879\text{V } (^{85}\text{Rb } F=3),$$

$$V_3 = -0.323340\text{V } (^{85}\text{Rb } F=2).$$

No matter how many changes were made to improve the situation in the experiment, dip 4 was not obtainable.

The voltage difference is therefore

$$\Delta V = |V_2 - V_1| = 0.03322 \text{ V}.$$

The value of Δv was calculated by measuring the distance between the dips in *Figure 3* which can be found in the manual.

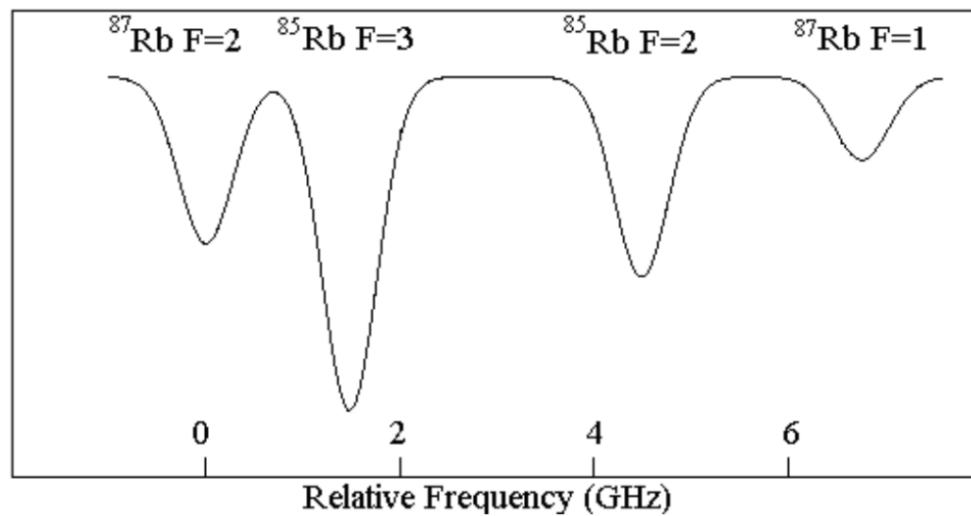


Figure 3 Doppler-broadened spectrum extracted from the manual

The calculated value is approximately

$$\Delta\nu \approx 1.48 \text{ GHz.}$$

This approximation has many sources of error, such as inaccuracies in the measurement between dips. Therefore, the result should be interpreted with caution.

The ratio $\frac{d\nu}{dV}$ is therefore:

$$\frac{d\nu}{dV} = \frac{1.48 \text{ GHz}}{0.03322 \text{ V}} = 44.55 \frac{\text{GHz}}{\text{V}}.$$

Since the frequency scan in this setup is accomplished by current tuning of the diode laser rather than piezo tuning, which usually produces a much stronger frequency response to voltage changes, the obtained value is reasonable. Typical current-tuned diode lasers show tuning sensitivities in the range of 3–150 GHz/V, which places our measured value comfortably within the expected region.

3.2 Plotting the voltage signals over frequency and identifying the different peaks

To obtain reliable frequency-resolved absorption spectra, the raw measurement data had to be processed in several steps. The voltage signal of Channel 2, which provides the scanning ramp for the laser frequency, exhibited strong noise and short-scale fluctuations that caused the direct voltage–frequency conversion to smear out the Doppler-free features almost completely. To address this, a linear fit was applied to the Ch2 data, effectively replacing the noisy ramp with its best linear approximation. This procedure removes high-frequency fluctuations while preserving the global frequency sweep. For this linearization, the statistical uncertainty of the fit was evaluated by calculating the RMS deviation between the measured Ch2 data and the fitted line. This RMS value was then propagated through the voltage-to-frequency calibration factor determined in the previous section, yielding the corresponding frequency uncertainty. After the ramp had been linearized, the fitted Ch2 values were converted into absolute frequency using the experimentally determined conversion factor (in units of MHz/V).

With the frequency axis stabilized, the photodiode signal of Channel 1 was plotted against frequency for each absorption dip. For every selected region, the code automatically detected local extrema, refined their positions, and extracted the central frequencies of the observable hyperfine peaks. In most regions only the most pronounced structures could be detected, which is probably due to the ramp noise, residual frequency jitter, and the limited signal-to-noise ratio of the photodiode. Because the conversion factor between ramp voltage and optical frequency is extremely large, even small fluctuations in the ramp signal translate into large frequency variations, further limiting the visibility of weaker hyperfine features.

The uncertainties in the extracted peak positions were obtained by propagating the previously calculated frequency uncertainty of the ramp linearization, providing a consistent error estimate for all measured hyperfine transitions.

	RMS [V]	frequency uncertainty [MHz]
Peak 1	$2.53 \cdot 10^{-3}$	113
Peak 2	$2.56 \cdot 10^{-3}$	115
Peak 3	$3.47 \cdot 10^{-3}$	154

Table 1 Calculated RMS and frequency uncertainty due to linearization of the ramp voltage

The resulting peak separations were then compared to the theoretical hyperfine and crossover transitions shown in *Figure 4*.

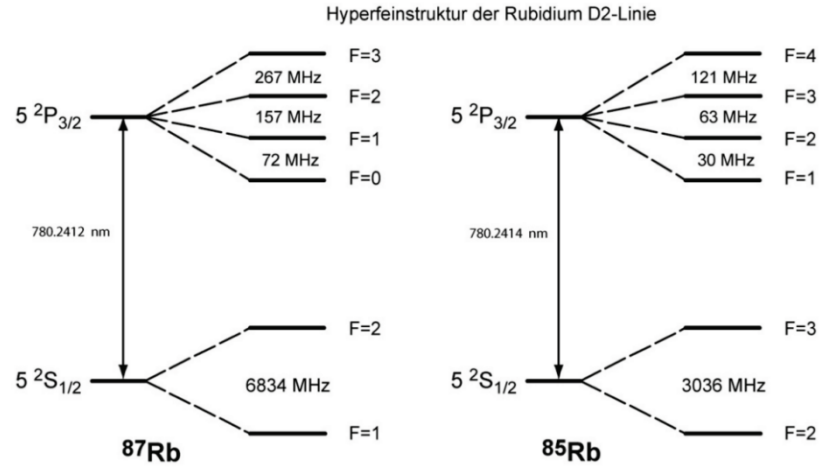


Figure 4 Energy-level diagram of the two Rubidium isotopes

3.3 Entire Doppler-broadened absorption spectrum plotted over frequency

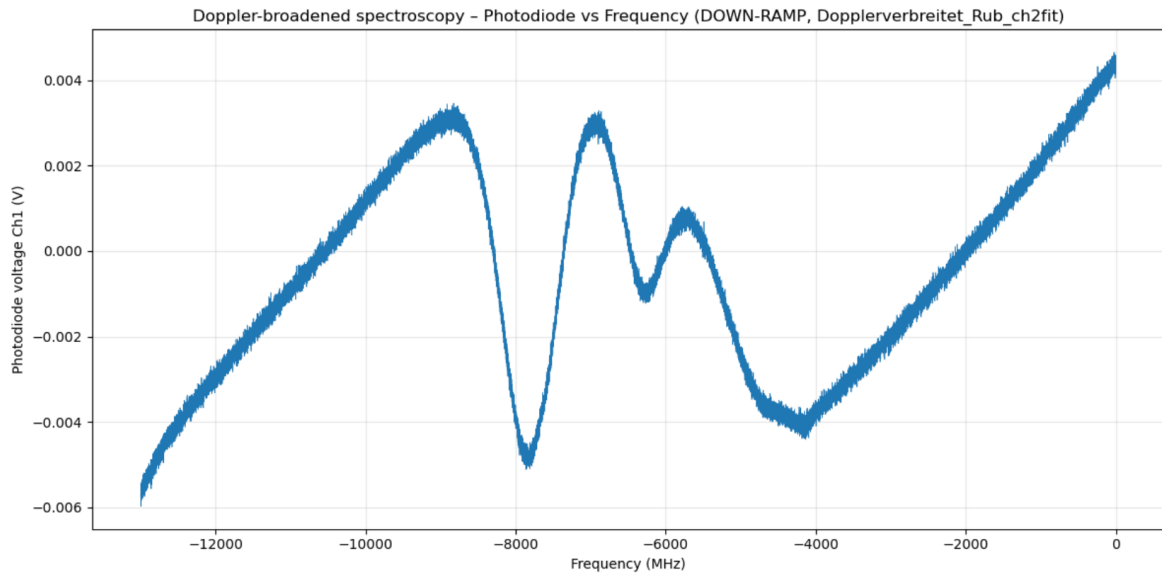


Figure 5 Entire Doppler-broadened absorption spectrum plotted over frequency

Figure 5 shows the Doppler-broadened absorption spectrum plotted over frequency.

The residual noise of this fit corresponds to an RMS voltage uncertainty of $\sigma_{\text{Ch2}} \approx 3.45 \times 10^{-3}$ V, which translates into a frequency uncertainty of $\sigma_f \approx 154$ MHz.

3.4 Entire Doppler-free absorption spectrum plotted over frequency

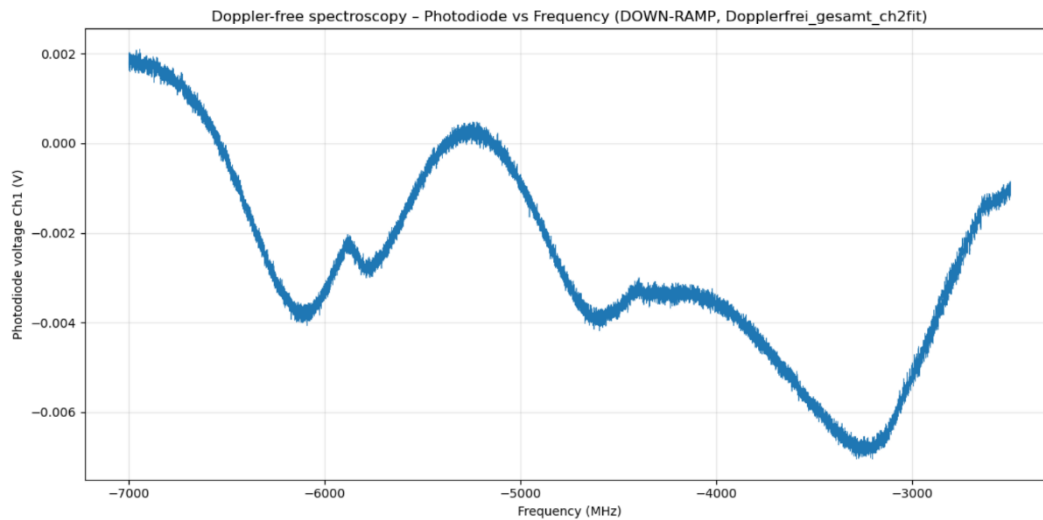


Figure 6 Entire Doppler-free absorption spectrum plotted over frequency

Figure 6 shows the Doppler-free absorption spectrum plotted over frequency in an interval of around 5 GHz.

Even though the frequency axis is obtained from a clean linear fit of Ch2 (with a frequency uncertainty of about 156 MHz), the strong fluctuations in the final spectrum come from the photodiode signal Ch1 itself. Ch1 contains high-frequency noise, and when this noisy signal is spread over a very large frequency range (several GHz), the noise becomes visually amplified. The peaks are still present, but they appear less pronounced because the intrinsic Ch1 noise dominates after the frequency rescaling.

3.4.1 Dip 1 (87Rb F=2)

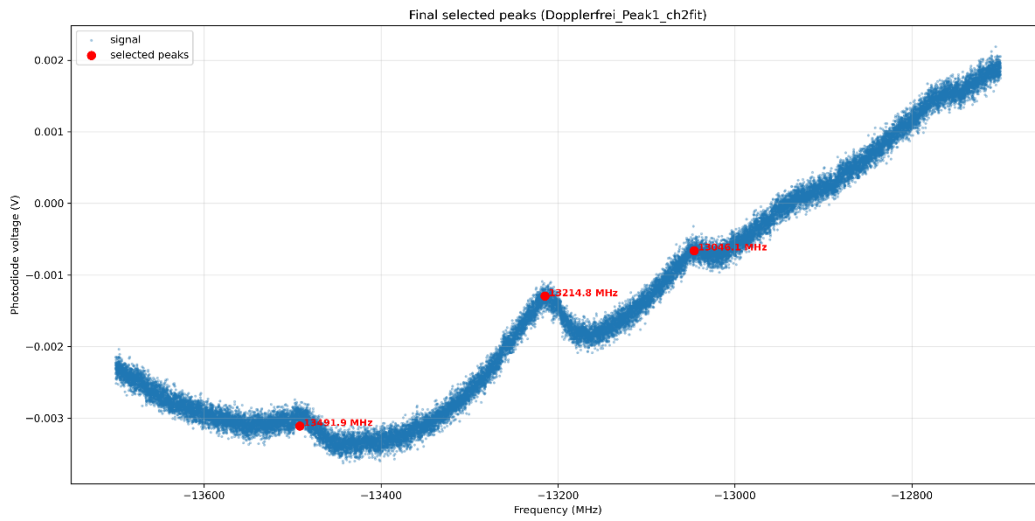


Figure 7 Identified peaks for 87Rb F=2

Peak	Measured $\Delta\nu$ [MHz]	Theoretical $\Delta\nu$ [MHz]	Deviation [MHz]	Deviation [%]	Assignment
Peak 1 (13.5GHz) $\leftrightarrow$ Peak 2 (13.2GHz)	277.1	267	10.1	3.8	$F = 3 \leftrightarrow F = 2$
Peak 2 (1.32GHz) $\leftrightarrow$ Peak 3 (13.0GHz)	168.7	157	11.7	7.5	$F = 2 \leftrightarrow F = 1$

Table 2 Identified peaks, their frequency differences, and the corresponding hyperfine transitions
(87Rb F=2)

The peak on the left therefore corresponds to F=3, the middle one to F=2 and the one on the right side to F=1. The values of the hyperfine peaks agree well with the theoretical values within the expected uncertainties.

No clear cross-over resonances were observed in this region, which is most likely due to the previously discussed restrictions.

3.4.2 Dip 2 (85Rb $F=3$)

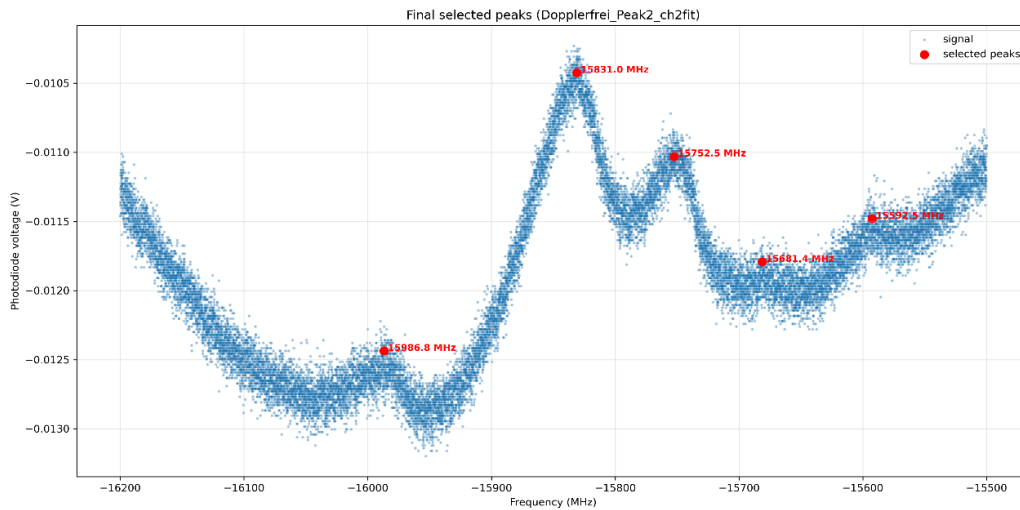


Figure 8 Identified peaks for 85Rb $F=3$

In this region, five local maxima were detected as expected for the three hyperfine transitions and their associated crossover resonances. The two central peaks can be unambiguously assigned to genuine hyperfine transitions, since both their shape and their frequency separation match the theoretical values very well.

However, the assignment of the remaining outer structures is ambiguous: the peak on the left has the characteristic shape and size of a true hyperfine resonance, while the rightmost peak also lies at a frequency difference that would be consistent with the expected hyperfine splitting. Depending on which of these two peaks is interpreted as the third hyperfine transition, the remaining peak would have to be classified as a crossover resonance - both interpretations are compatible with the theoretical level scheme and the measured frequency spacings.

To solve this problem, the analysis will contain both cases.

3.4.2.1 Case 1: left peak is a hyperfine peak

Peak	Measured $\Delta\nu$ [MHz]	Theoretical $\Delta\nu$ [MHz]	Deviation [MHz]	Deviation [%]	Assignment
Peak 1 (15.9GHz) $\leftrightarrow$ Peak 2 (15.8GHz)	155.8	121	34.8	28.7	$F = 4 \leftrightarrow F = 3$
Peak 2 (15.8GHz) $\leftrightarrow$ Peak 3 (15.7GHz)	78.5	63	15.5	24.6	$F = 3 \leftrightarrow F = 2$

Table 3 Identified peaks, their frequency differences, and the corresponding hyperfine or crossover transitions for case 1 (85Rb $F=3$)

3.4.2.2 Case 2: right peak is a hyperfine peak

Peak	Measured $\Delta\nu$ [MHz]	Theoretical $\Delta\nu$ [MHz]	Deviation [MHz]	Deviation [%]	Assignment
Peak 1 (15.8GHz) $\leftrightarrow$ Peak 2 (15.7GHz)	78.5	63	15.5	24.6	$F = 2 \leftrightarrow F = 3$
Peak 2 (15.7GHz) $\leftrightarrow$ Peak 4 (15.5GHz)	160	121	39	32.2	$F = 3 \leftrightarrow F = 4$
Peak 2 (15.7GHz) $\leftrightarrow$ Peak 3 (15.6GHz)	71.1	60.5	10.6	17.5	$F = 3 \leftrightarrow$ Cross-Over-Peak between $F = 3$ and $F = 4$

Table 4 Identified peaks, their frequency differences, and the corresponding hyperfine or crossover transitions for case 2 (85Rb $F=3$)

Purely based on the deviation values, case 1 is most likely the correct assignment.

3.4.3 Dip 3 (85Rb $F=2$)

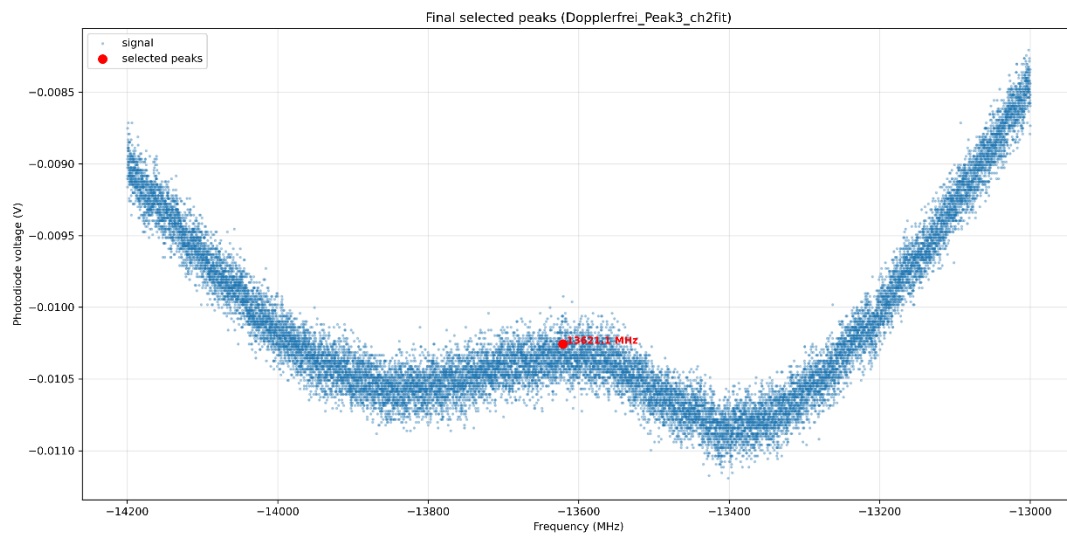


Figure 9 Identified peaks for 85Rb $F=2$

For this transition only a single hyperfine feature was visible, which made the analysis and identification of the peaks impossible. The reason for that is probably that the data in this region is significantly degraded by noise.

3.5 Weighted triplet fitting

The three Doppler-broadened absorption dips observed in the rubidium spectrum correspond to the transitions (see last chapter):

$$^{87}\text{Rb } (F = 2 \rightarrow F')$$

$$^{85}\text{Rb } (F = 3 \rightarrow F')$$

$$^{85}\text{Rb } (F = 2 \rightarrow F')$$

Each of these resonances consists of three closely spaced hyperfine transitions with different transition probabilities.

Therefore, each dip was fitted with a weighted triplet of Gaussian functions sharing a common full width at half maximum (FWHM) and baseline.

The relative weights of the three components were chosen according to the Clebsch–Gordan coefficients of the hyperfine transitions listed in the manual:

For $^{87}\text{Rb } (F = 2)$ the ratio (1 : 5 : 7), and for $^{85}\text{Rb } (F = 3)$ the ratio (10 : 35 : 81) was used.

A global non-linear least-squares fit was performed for each dip using a custom Python routine.

The fitted regions were manually selected to exclude overlapping crossover resonances.

After calibration of the frequency scale using the known isotope separation (^{85}Rb splitting = 3035.732 MHz), the corrected parameters were obtained as:

Dip	FWHM [MHz]	σ FWHM [MHz]
$^{87}\text{Rb } F = 2$	490.0	101.1
$^{85}\text{Rb } F = 3$	397.0	50.6
$^{85}\text{Rb } F = 2$	532.3	123.1

Table 5 FWHM for each dip

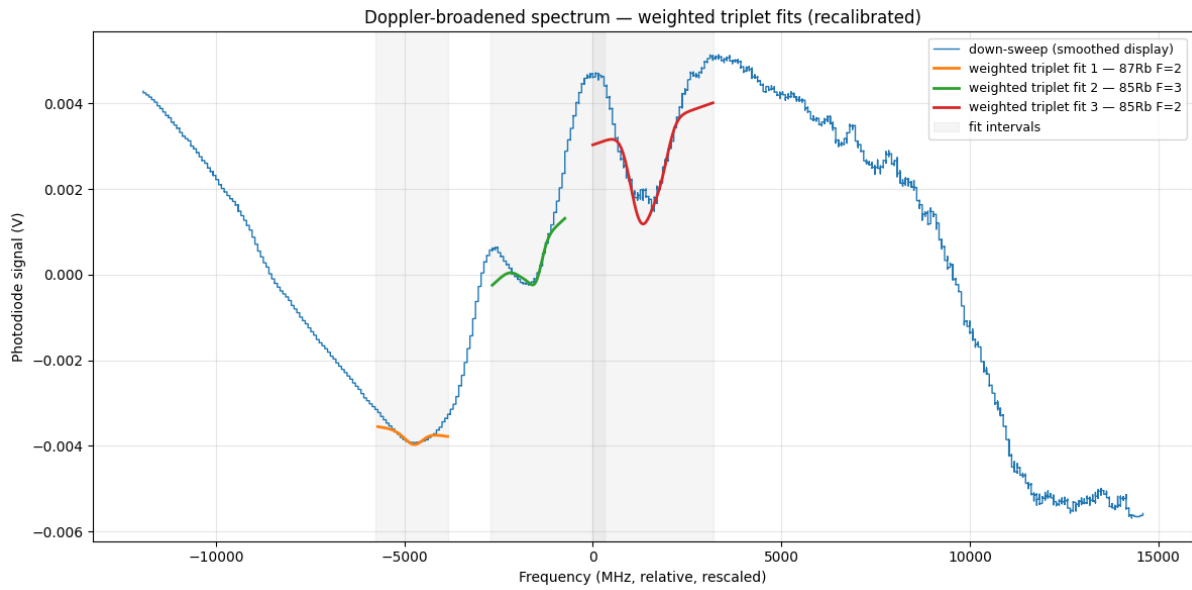


Figure 10 Gaussian fits for the Doppler-broadened spectrum

The resulting weighted triplet fits are shown in *Figure 10*.

All fits reproduce the observed dips well, including the weaker right-hand resonance with minor crossover interference.

3.6 Determination of temperature and discussion

The Doppler width of an absorption line in a gas is directly related to the thermal velocity distribution of the atoms and thus to the vapor temperature.

Assuming purely Gaussian broadening, the relation between the measured FWHM and the temperature is given by

$$T = \frac{M}{8k_B \ln(2)} \left(\frac{\Delta\nu_D}{\nu_0} c \right)^2,$$

where M is the atomic mass of rubidium, k_B is the Boltzmann constant,

ν_0 the optical transition frequency (≈ 384 THz for 780 nm), and $\Delta\nu_D$ the measured Doppler width.

Using the experimentally determined FWHMs from *Table 1* and their standard deviations as uncertainty, the temperatures and most probable atomic velocities were calculated as:

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Dip	$T[\text{K}]$	$v_w[\text{m/s}]$
$^{87}\text{Rb } F = 2$	275.5	229.6
$^{85}\text{Rb } F = 3$	176.7	186.0
$^{85}\text{Rb } F = 2$	317.7	249.4

Table 6 Temperature and velocity for each dip

The weighted mean temperature over all three dips is therefore

$$\langle T \rangle = (256.6 \pm 72.3) \text{ K}$$

$$\langle v_w \rangle = (221.7 \pm 32.4) \text{ m/s.}$$

3.7 Discussion of uncertainties and deviations

The obtained mean temperature is slightly below room temperature but well within the expected range for the experiment. This deviation can be explained by several systematic effects:

1. **Actual cell temperature:** The measurement was performed in a laboratory with active air conditioning. The rubidium cell was therefore at an ambient temperature of approximately 20°C ($\approx 293 \text{ K}$) or slightly lower, which is consistent with the derived value of $(256.6 \pm 72.3) \text{ K}$.
2. **Frequency calibration uncertainty:** The conversion factor from voltage to frequency depends on the linearity of the ramp. A small deviation in slope ($\approx \pm 5\%$) directly scales into the Doppler width and thus into T .
3. **Crossover resonance interference:** In the case of the $^{85}\text{Rb } F = 2$ resonance (Dip 3), a small crossover peak was observed within the Doppler-broadened region. To prevent distortion of the Gaussian fit, the affected frequency interval around the crossover resonance was excluded from the fitting range. As a result, the determined FWHM and corresponding temperature for this dip carry a somewhat higher uncertainty compared to the other two resonances.

Taking all these effects into account, the derived temperature agrees well with the expected ambient conditions and validates the accuracy of the Doppler-broadening analysis.

3.8 How to improve the frequency calibration

By employing an absolute frequency reference rather than depending solely on the laser scan's voltage-to frequency slope, the frequency calibration can be greatly enhanced. For instance, evenly spaced transmission peaks with a known frequency spacing would result from connecting a Fabry-Perot-interferometer (or an etalon with a known free spectral range) in parallel with the spectroscopic signal. Any scan nonlinearity could be corrected which would greatly lower the MHz scale's uncertainty.

3.9 Frequency stabilization and relative accuracy of the diode laser

Using saturation spectroscopy, the frequency of the diode laser can be stabilized with an accuracy of approximately the natural line width (≈ 6 MHz). Given the absolute laser frequency of about 3.84×10^{14} Hz, this corresponds to a relative accuracy of roughly 1.6×10^{-8} or 0.0000016%.

$$\Delta\nu \approx 6 \text{ MHz}$$

$$\nu_0 = \frac{c}{\lambda} = \frac{3 \times 10^8}{780 \times 10^{-9}} \approx 3.84 \times 10^{14} \text{ Hz}$$

$$\frac{\Delta\nu}{\nu_0} = \frac{6 \times 10^6}{3.84 \times 10^{14}} \approx 1.6 \times 10^{-8}.$$

3.10 Conclusion

In this experiment, the Doppler-broadened and Doppler-free absorption spectra of rubidium were successfully recorded and analyzed. From the Doppler-broadened spectrum, the frequency–voltage calibration factor of the laser scan was determined, leading to a frequency axis with an uncertainty of about 100–150 MHz. Using weighted Gaussian fits for the three Doppler dips, the Doppler widths were obtained and used to calculate the temperature of the atomic vapor. The resulting mean temperature,

$$\langle T \rangle = (256.6 \pm 72.3) \text{ K}$$

shows a large uncertainty. Although the central value is below room temperature, the uncertainty range still includes the expected ambient temperature.

In the Doppler-free saturation spectroscopy, the hyperfine structure of rubidium was partially resolved. For ^{87}Rb ($F = 2$) all three hyperfine transitions were identified, and their frequency separations agreed with theoretical values within the experimental uncertainty. For ^{85}Rb ($F = 3$) several hyperfine and crossover resonances were observed. Although noise and ramp fluctuations introduced ambiguities in the assignment of some outer peaks, the measured separations remained consistent in magnitude with the expected hyperfine splittings. For ^{85}Rb ($F = 2$), only one peak was visible due to strong noise, which made a full analysis impossible.

Across all measurements, the main limitation was the instability of the frequency scan. The ramp signal had to be linearized, and remaining noise produced a significant frequency uncertainty. Nevertheless, the key spectral features such as Doppler dips, hyperfine transitions and crossover resonances were clearly detected. The weighted triplet fits reproduced the Doppler-broadened profiles well and confirmed the validity of the fitting procedure.

Overall, the experiment demonstrated Doppler-broadened and Doppler-free spectroscopy, confirmed the expected hyperfine structure of rubidium, provided a realistic temperature estimate of the atomic vapor, and showed that saturation spectroscopy enables frequency stabilization at the MHz level. Despite limitations due to scan nonlinearity and noise, the results agree well with the theoretical predictions and support the correctness of the experimental procedure and the data analysis.

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